



DSE6311 - Capstone - Module 01

Project Ideas and Proposals

Objectives

- Understand what is expected from you
- Find your team members and choose your roles
- Develop a data science capstone question
- Construct a preliminary proposal for feedback
- Learn or remember how to set up a GitHub repo

To-Do List

1. **Set up your team's GitHub repo, following all guidelines for full-credit**
2. **Work with your team to develop your Preliminary Project Proposal**
3. **Complete an Individual Development Plan (IDP)**



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Mathematics
BS

Industrial Eng.
PhD

Thermal Analytics
Sr Research Engineer

Data Science
Team Lead and Manager

Load Forecasting
Lead Quantitative Analyst

Data Science
Adjunct Lecturer

2001-05

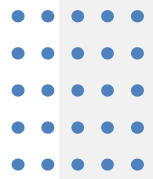
2005-10

2010-18

2018-21

2021-

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What is the
goal of this
course
about?

You will build and work on a data science problem from start to finish, applying the skills and knowledge you have acquired from multiple classes.

Course Learning Objectives

- Apply data science methodologies to solve a complex, real-world problem.
- Collect, clean, and preprocess data for analysis.
- Develop and evaluate machine learning models for predictive or classification tasks.
- Communicate findings and insights through reports and presentations.
- Work effectively in a team, if applicable, to complete a collaborative project.



Forming Teams

- Is it required to join a team?
 - No, but I **strongly** recommend it for your sanity.
- Find teammates with similar interests, backgrounds, or coding languages.
- Join a team by **Saturday October 25**.
 - You must join a “team” even if you are working solo, and please send me an email.
- Your preliminary project proposal is already due **Tuesday, October 29**!

Team structure

- **Team Lead**

- organize the team & delegate data management or coding tasks
- make decisions (diplomatically not unilaterally) & handle disputes
- guide and keep track of the team's progress
- **post team weekly deliverables**

- **Recorder**

- maintain documentation (code & deliverables)
- do not do all the writing or documentation
- organize & delegate writing tasks
- **post team weekly stand-up discussion post**

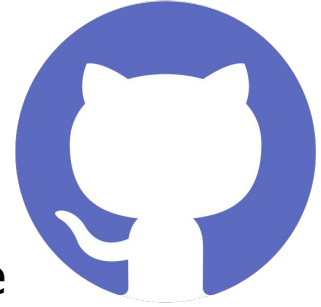
- **Spokesperson**

- be the first to speak for the group when asked
- organize presentations and delegate tasks
- do not prepare or deliver all presentations alone

Team peer review rubric (end of Capstone)

Criterion		Score			
	4 (Exceptional)	3 (Great)	2 (Average)	1 (Below Average)	0 (Poor)
Contributes to Team Meetings	Helps the team move forward by articulating the merits of alternative ideas.	Offers alternative solutions or courses of action that build on others' ideas.	Offers new suggestions to advance the work of the group.	Shares ideas but does not advance the work of the group.	Did not share ideas or advance the work of the group.
Facilitates the Contributions of Team Members	Engages team members to facilitate their contributions to meetings. Constructively builds upon or synthesizes the contributions of others. Notices when someone is not participating and invites them to engage.	Engages team members in ways that facilitate their contributions to meetings by constructively building upon or synthesizing the contributions of others.	Engages team members in ways that facilitate their contributions to meetings by restating the views of other team members and/or asking questions for clarification.	Engages team members by taking turns and listening to others without interrupting.	Did not engage with team members, interrupted consistently, or did not show respect to others.
Individual Contributions Outside of Team Meetings	Completes all assigned tasks by deadline; work accomplished is thorough, comprehensive, and advances the project. Proactively helps others complete their assigned tasks to similar excellence.	Completes all assigned tasks by deadline; work accomplished is thorough, comprehensive, and advances the project.	Completes all assigned tasks by deadline; work accomplished advances the project.	Completes all assigned tasks by deadline.	Did not complete assigned tasks by deadline.
Fosters Constructive Climate	<i>Supports a constructive team climate by doing all of the following:</i>	<i>Supports a constructive team climate by doing any three of the following:</i>	<i>Supports a constructive team climate by doing any two of the following:</i>	<i>Supports a constructive team climate by doing any one of the following:</i>	Fails to support a constructive team climate by doing none of the following:
	Treats team members respectfully using polite and constructive in communication.	Treats team members respectfully using polite and constructive in communication.	Treats team members respectfully using polite and constructive in communication.	Treats team members respectfully using polite and constructive in communication.	Treats team members respectfully using polite and constructive in communication.
	Engages with other team members in a cooperative rather than divisive manner.	Engages with other team members in a cooperative rather than divisive manner.	Engages with other team members in a cooperative rather than divisive manner.	Engages with other team members in a cooperative rather than divisive manner.	Engages with other team members in a cooperative rather than divisive manner.
	Motivates teammates by expressing confidence about the importance of the task and the team's ability to accomplish it.	Motivates teammates by expressing confidence about the importance of the task and the team's ability to accomplish it.	Motivates teammates by expressing confidence about the importance of the task and the team's ability to accomplish it.	Motivates teammates by expressing confidence about the importance of the task and the team's ability to accomplish it.	Motivates teammates by expressing confidence about the importance of the task and the team's ability to accomplish it.
	Provides assistance and/or encouragement to team members.	Provides assistance and/or encouragement to team members.	Provides assistance and/or encouragement to team members.	Provides assistance and/or encouragement to team members.	Provides assistance and/or encouragement to team members.
Responds to Conflict	Addresses destructive conflict Directly and constructively assists to resolve conflict in a way that strengthens team cohesiveness and effectiveness.	Identifies and acknowledges conflict and stays engaged with it.	Redirecting focus toward common ground, toward task at hand (away from conflict).	Passively accepts alternate viewpoints/ideas/opinions.	Does not respond to conflict or directly contributes to conflict.

Use GitHub to organize your project



- If you do not yet have a GitHub account of your own, you will need to **make one**
- If you are very comfortable with git on the command line, I encourage you to use **GitHub Desktop** instead
- Please reach out to me via e-mail if you need help with GitHub. **Happy to help!**
- By **Tuesday**, you should:
 - (1) make your repo
 - (2) each teammate should be on the repo as a collaborator
 - (3) you should have added me to your repo as a collaborator
 - **One person will be the owner of your GitHub repo.**

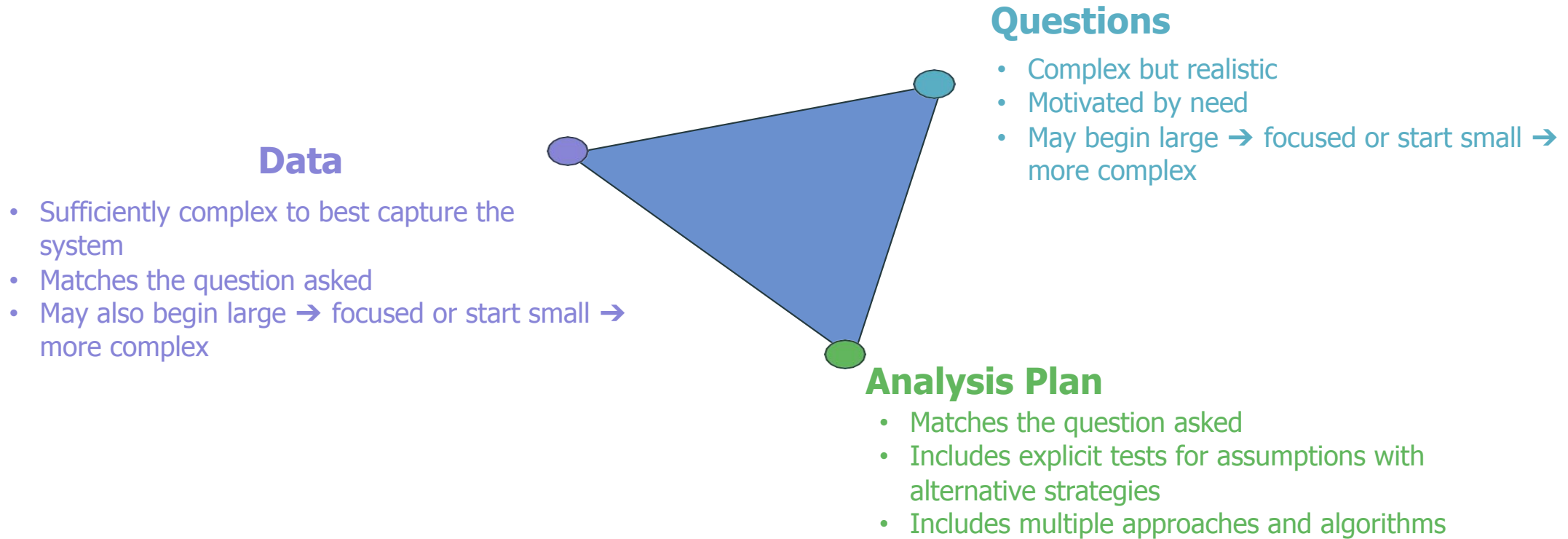
What makes a great research question?

Complex questions about how the world or some subset of the world (e.g., industry, policy) works

- Do Parkinson's disease rates correspond to soil organophosphate (pesticide) levels?
- Have hate crimes escalated in the U.S. and, if so, why?
- Do employees with post-baccalaureate degrees have lower unemployment rates?

Complex questions often require complex data sources (or more than one data source).

What leads to a great research outcome?



Choosing your question: **improvement route**

Scenario: *Octavia decided to join this [Kaggle competition](#), but her model underperformed. In fact, she had serious overfitting due to variance inflation.*

What new variables or analysis (maybe disallowed by the competition) could Octavia choose?



EESTI ENERGIA · FEATURED CODE COMPETITION · 2 MONTHS TO GO

Enefit - Predict Energy Behavior of Prosumers

Predict Prosumer Energy Patterns and Minimize Imbalance Costs.

[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#)

Dataset Description

Your challenge in this competition is to predict the amount of electricity produced and consumed by Estonian energy customers who have installed solar panels. You'll have access to weather data, the relevant energy prices, and records of the installed photovoltaic capacity.

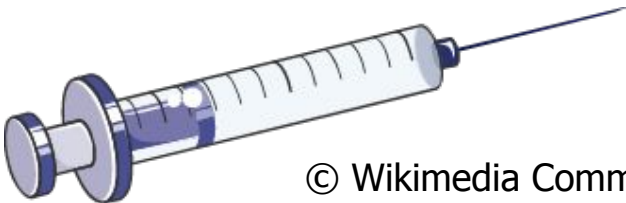
This is a forecasting competition using the time series API. The private leaderboard will be determined using real data gathered after the submission period closes.



Choosing your question: **novel route**

- **Scenario:** *Luan works for his state health board, and he has a new project at work identify social determinants of parents' decisions to vaccinate their children. Luan lives and works in a very socially and culturally heterogeneous state, so he decides to analyze this at the nationwide level to boost his sample size.*
 - *Young child vaccination rates ([CDC](#))*
 - *Poverty & access to internet, including via smartphones ([Agency for Healthcare Research & Quality](#))*
 - *U.S. Census data on county demographics (race, ethnicity, household size, median household income)*

What else could Luan choose to include?



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Use original data sources.

- Even if inspired by a competition or other analysis, you are required to return to the **original dataset** and practice cleaning, processing, and otherwise readying your data for analysis.
- Original datasets are important because they may be refreshed, corrected, or improved
- Citations are important. (Citation style, however, isn't.)

How do we develop a good capstone project question?

Unique	Prospective employers want to see that you can think independently and “outside the box” – even if they will ultimately give you data to analyze
Specific & Focused	Your question, hypotheses, predictions, objectives, & scope are clear, relevant, & appropriate. KNOW YOUR STAKEHOLDER!
Attainable	Our questions can be <u>too</u> narrow, alas; we are seeking the “sweet spot” of attainability
Unbiased	Your job as a data scientist is to be as objective and unbiased as possible: you do not know the outcome.
Evaluable	Design your analysis with a clear metric for evaluation in mind. You must have an analysis plan in mind as you develop your question!
Valuable & Useful	Who will find this a valuable use of your time? Who will fund this? Why will anyone care? KNOW YOUR STAKEHOLDER!
Timely	It must be achievable within the project timeline (6-7 weeks, in our case). Timeliness also matters – is it relevant to what is happening now?
Complex	The best models have sufficient variation, parameters, and samples to train. Complexity is both a bane & a boon!

Actionable things we can do to prepare a **stellar** proposal:

Do Research

We will often have to analyze something about which we know little – which is both the cool thing and the challenge about being a data scientist!

Create an Analysis Plan

The analysis plan **must** match the question, objectives and aim of the project, and is often one of the hardest aspects to preparing the proposal.

Anticipate Disaster

Assess the landscape of your question, including:

Barriers

Risks

Pitfalls

Privacy issues

Governance issues

Project
Management

Acquire the Data

Remember **GIGO**: Garbage In, Garbage Out. What cleaning issues are there to include in your analysis plan? What encoding needs to be done? How to handle the missing values? Should you include transformations of any kind in your analysis plan?

Clean the Data

Hypothesis vs. Prediction

You are not research scientists – but you are data scientists

So, you still need to generate clear hypotheses and predictions!

- **Hypothesis** – explanation of how / why something works (“because”)
- **Prediction** – explanation of what you expect to happen (backed by motivation)

Hypothesis vs. Prediction

Estonia energy prediction case:

- Hypothesis: Prosumer data can be used to predict energy consumption data **because** it tells us about their patterns of energy used relative to current environmental conditions.
- Prediction: Prosumers are more likely to be in consumption mode when environmental conditions are unfavorable to production, after accounting for high-use or low-production cases.

Original vaccination case:

- Hypothesis: Young child vaccination rates may be affected by household internet access **because** both internet connectivity could alter the flow of information about vaccines and vaccine availability for young children to their parents.
- Prediction: Households with higher internet connectivity, rather than low, after correcting for sociodemographics, are more likely to withhold young childhood vaccines because of ready availability of misinformation through the internet.

But First... Preliminary Proposal (Due October 29)

Background & Question

- A defined research question that serves a need or fills a niche
 - What is the question?
 - What need or niche does it fill?
 - Why is it worth your time/effort to explore this question?
 - Is your question novel / original? If not, why not? (**You might be taking the improvement approach.**)
- A hypothesis and prediction. These are not the same thing.

**Taken directly from your
assignment details in
Canvas.**

Data & Analysis

- What data set(s) have you found that you think are a good match for your question. Why?
 - Make sure to provide a link to a data source.
- What response / outcome variable will you use?
- What predictor variable(s) will you use?
- What is your tentative analysis plan?
 - Are there any pitfalls you can see with this plan? Anything that could topple your idea?
- How will you know if your question is answered?
- How will you know if your hypothesis is supported?

Final Project Proposal Rubric (Due November 5)

Criteria		Score			
	4 (Exceptional)	(Great)	2 (Average)	1 (Below Average)	0 (Poor) *
Background Motivation	Background context is provided that sets the stage for the questions and analyses to follow. Background is well- researched and thoroughly grounded in theory, policy, business practices, or other relevant context. Citations are provided. At least one paragraph is provided.	Background context is provided that sets the stage for the questions and analyses to follow. Background is researched but citations are inappropriate or the background is mismatched to the question proposed. At least one paragraph is provided.	Background context is provided but is not appropriately well-researched or grounded in relevant context. Citations may be missing. Or, less than one paragraph is provided.	Background context is attempted but has two or more of the following issues: is inappropriate, was not researched, lacks citations, or is under one paragraph.	No background context is provided or it is incoherent, plagiarized, or obviously AI-generated.
Questions	Questions are well-motivated, interesting, insightful, and novel. Questions are supported by references to highlight need in industry, government, or academia.	Questions are novel, appropriate, coherent, and motivated. Questions are grounded in theory or motivated by need, but lacks proper references or support are unclear or inappropriate.	Questions are not novel, appropriate, coherent, or motivated. Questions may be novel or coherent, but lack motivation of support. Alternatively, references may be provided but the question is not novel or coherent.	Questions are not novel or are overly simplistic, unrelated, or unmotivated. They lack any kind of grounding in theory or need-based motivation.	No questions are provided or are incoherent or obviously plagiarized.
Hypothesis & Prediction	Hypothesis and predictions are clear and appropriate. Provides written support for why the hypotheses and predictions are appropriate to the questions.	Hypothesis and predictions are clear and appropriate. Provides written support for hypotheses and predictions, but they are mismatched to the questions.	Hypotheses and predictions are attempted but may be inappropriate or unclear. Or, hypotheses and predictions are provided but written support is missing.	Hypothesis and predictions are unclear or inappropriate. Written support is missing.	No hypotheses or predictions provided.
Proposed Data Sources	Data chosen are excellently matched to answer the questions and hypotheses proposed. Data sources are provided. It is clear how the original data were collected, and if there are any concerns or caveats in the data collection methods.	Data chosen are a good match to answer the questions and hypotheses posed. Data sources are provided. An attempt to explain how the original data were collected is provided but concerns or caveats may be missing.	Data chosen are a reasonable match to answer the questions and hypotheses posed. Data sources are provided. An attempt is made to explain the original data collection methods, but may be incomplete or mistaken. Concerns or caveats may be missing.	Data chosen are a poor match to answer the questions and hypotheses posed. Data sources are provided. Explanations of original collection methods and/or caveats or concerns are missing.	Data chosen are an unreasonable match to answer the questions and hypotheses posed or data sources are not provided. Or, it is obvious that the instructor or peers have tried to help the team find more appropriate data sources but were ignored.
Meets Requirements, Syntax, & Mechanics	Student exceeds stated requirements. Student goes above and beyond to demonstrate exceptional engagement. Student uses skillful and accurate language that communicates meaning to readers with clarity. Writing is in complete sentences and is virtually error-free.	Student meets stated requirements. Student demonstrates engagement. Student uses straightforward language that generally conveys meaning to readers. Writing is in complete sentences and is virtually error-free.	Student meets stated requirements but does not demonstrate engagement. Student uses language that generally conveys meaning to readers with clarity, although writing may include some errors or be incomplete sentences.	Student does not meet stated requirements or demonstrate engagement. Uses language that sometimes impedes meaning because of errors in usage.	Student does not meet stated requirements or demonstrate engagement. No attempt to correct errors seems to have been made or no writing is so full of errors that it is uninterpretable.

* Any writing that is obviously AI-generated or plagiarized will receive an automatic 0 across the entire assignment.

For everyone's reference, the program has compiled a great list of data repos and sites in Google Sheets to help you with your project proposals. These have been used by past capstone projects (and by myself personally in classes and work).

Please consider using this resource!

<https://docs.google.com/spreadsheets/d/19GhgorAl4FDKw9xAYShPCHJHI9KHuV1I5NqRpSogRRY/edit?usp=sharing>